MILITARY SPECIFICATION

BURET, OXYGEN

1. SCOPE

1.1 This specification covers one type of oxygen buret.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

DD-G-541 Glass (Laboratory).

STANDARD

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

DRAWING

Defense Medical Supply Center

20889 Buret, Oxygen.

FSC 6640

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. DEPARTMENT OF COMMERCE-NATIONAL BUREAU OF STANDARDS

Commercial Standard CS21-58	Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.
NBS Circular 602	Testing of Glass Volumetric Apparatus.

(Application for copies, accompanied by money order, coupon, or check, should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402. Price may be obtained from the Superintendent of Documents.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation C148-59T	Tentative Methods of Polariscopic Examinatio of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material. Glass shall be borosilicate glass conforming to DD-G-541, type I, class A or B.

3.2 Construction. Buret shall be constructed in accordance with Drawing 20389. Any cross section of the graduated portion of the inlet and outlet tubes taken in a plane perpendicular to their longitudinal axes shall be circular. Thickness of walls of the graduated portion and inlet and outlet tubes shall be as specified on the drawing; where not specified, thickness shall be 1.6 ± 0.4 millimeters (mm.). Dimensions and shape of stopcock shall be in accordance with Commercial Standard CS21-58, stopcock designation 2, double oblique bore.

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3.3 Marking. Markings shall be permanent and shall be engraved, etched, and filled with pigment or stained and fired in. The appearance of filled, etched, and stained markings shall not change when tested as specified in 4.4.1. Engraved markings need not be tested.

3.3.1 Graduation markings. Graduation markings shall be sharply defined lines of uniform width (maximum 0.3 mm.). Graduation markings shall be perpendicular to the vertical axis of the buret and parallel to each other. Graduations shall be legibly numbered, as shown on the drawing.

3.3.2 Identification marking. Burets shall be legibly and permanently marked with the manufacturer's name or registered trademark.

3.4 Annealing. Annealing of buret shall be such that there is no severe strain. When tested as specified in 4.4.2, ring strain shall not exceed a No. 1 temper, and any longitudinal strain shall be faint and diffused.

3.5 Capacity accuracy. When tested as specified in 4.4.3, accuracy shall be as shown on the drawing.

3.6 Leakage. Stopcock shall not leak, when tested as specified in 4.4.4.

3.7 Workmanship. Workmanship shall be first class throughout. Burets shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	II	2.5
Major B defects	S-2	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.6

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the buret.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Permanency of marking	3.3	4.4.1	S-2*	**
Strain	3.4	4.4.2	S-4	**
Capacity	3.5	4.4.3	S-1	**
Leakage	3.6	4.4.4	S-1	**

* Sample shall be divided by three for separate performance of tests described in 4.4.1.1, 4.4.1.2.1, and 4.4.1.2.2.

** Acceptance number shall be zero.

4.3 Examination. The burets shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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Categories and defects*

Major A

101 Component part missing.
102 Buret not free of sharp edges.
103 Buret not free of cracks.
104 Buret not free of chips.
105 Buret not free of protrusions.
106 Graduation, capacity, calibration, and taper size markings not complete, correct, or legible.

Major B

151 Lubricated stopcock does not turn in barrel without binding.

Minor

201 Buret not free of bubbles.
202 Buret not free of scratches.
203 Manufacturer's name or registered trademark not complete, correct, or legible.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The burets shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Permanency of marking.

4.4.1.1 Resistance to alkali. The buret, with stopcock removed, shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.1.2 Resistance to acid.

4.4.1.2.1 Hydrochloric acid. The buret shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

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4.4.1.2.2 Sulfuric acid. The buret shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.2 Strain. Buret shall be tested for strain in accordance with ASTM Designation C148-59T, method A or B.

4.4.3 Capacity. Determination of capacity accuracy shall be made by measuring the amount of water delivered at 20° centigrade. A minimum of one graduation of each area of the buret (0 to 2, 2 to 10, 10 to 91, 91 to 99, and 99 to 100) shall be tested. Measurement of water shall be made gravimetrically or volumetrically. Gravimetric measurement shall be in accordance with the method described in NBS Circular 602, and measuring device used for volumetric measure shall be calibrated in accordance with NBS Circular 602.

4.4.4 Leakage. Leakage through the stopcock shall be tested with no lubricant on the barrel and plug. Plug shall be wetted with water and inserted in the barrel in the open position and turned 90° to a firm set. A vacuum of 25 inches of mercury shall be applied across the stopcock for a period of 1 minute. Evidence of leakage shall be the movement of liquid, possibly accompanied by the appearance of dry areas on the ground surfaces.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Level C. The commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Interior packages. Each package shall be marked to identify contents.

5.2.2 Exterior containers. Exterior containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract or purchase order number.

6. NOTES

6.1 Ordering data. Procurement documents should specify the number and date of this specification.

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6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
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Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-B-36361 (DM)

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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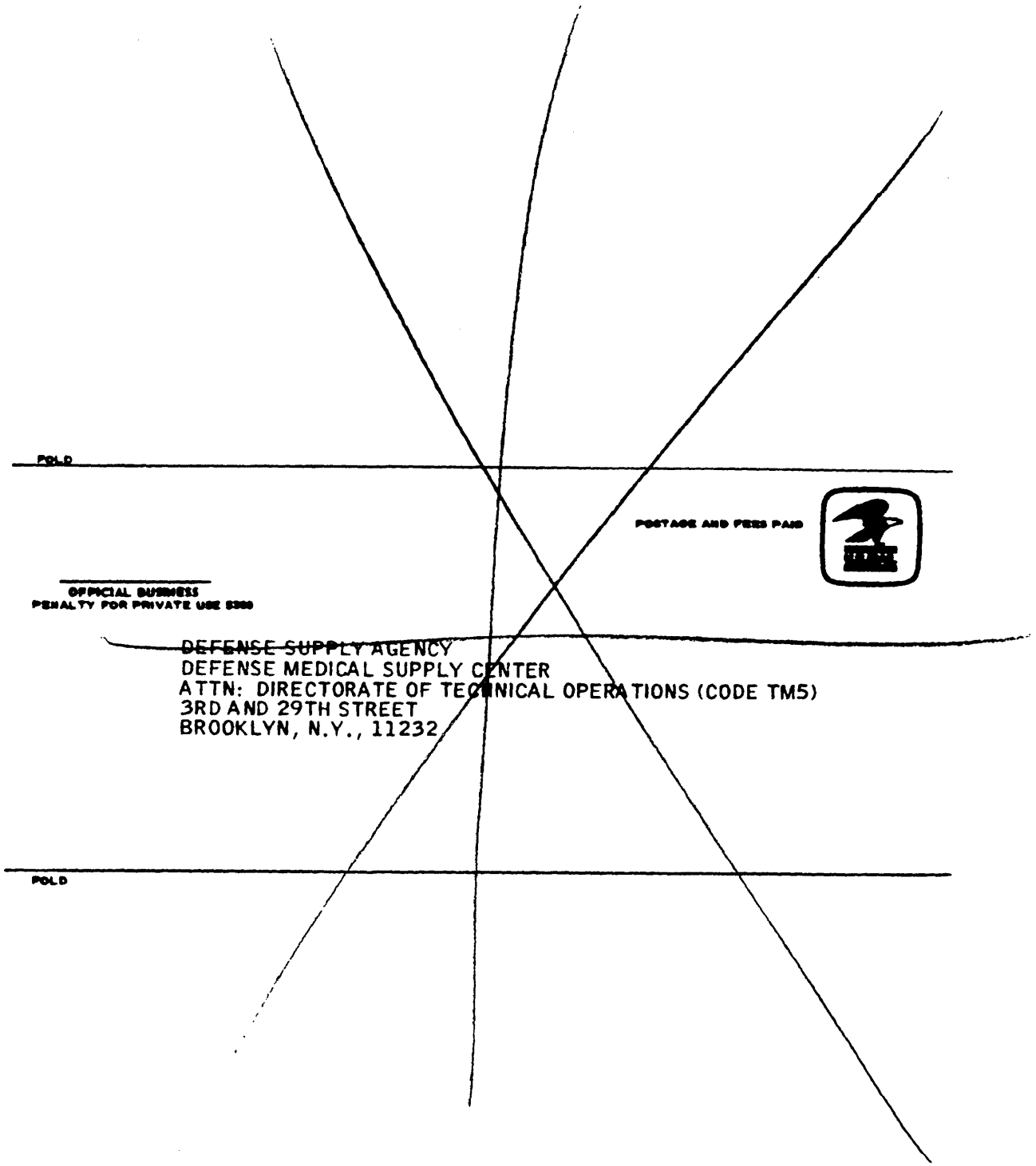
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